GRADE 8

Computational Thinking

National Level Examination

SUBJECT CODE			

Total Questions: 50 **Total Marks: 50** **Time: 75 Minutes**

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED TO DO SO

- **Carefully Check Your Name, Class, Roll Number, And Other Required Details On The OMR Answer Sheet.**
- **Do Not Forget To Sign The OMR Sheet. Also Write Your Roll No. On The Question Paper Booklet.**
- **Use Only Black/Blue Ball Pen And Hb Pencil To Fill The OMR Answer Sheet.**
- **All Questions Are Compulsory And Carry One Mark Each.**
- **Section A: Abstract & Spatial Thinking (10 Questions); Section B: Pattern Recognition (10 Questions); Section C: Decomposition, Data & Constraints (10 Questions); Section D: Algorithmic Thinking & Logical Flow (10 Questions); Section E: Ai Literacy & Olympiad Challenge (10 Questions).**
- **Only OMR Answer Sheets Will Be Evaluated.**
- **No Negative Marking For Incorrect Answers.**
- **You May Use The Space Provided On The Last Page For Rough Work (If Needed).**
- **Digital Devices Like Calculators, Smart Watches, Or Mobile Phones Are Not Allowed.**
- **After The Exam, Submit Both The OMR Sheet And The Question Paper To The Invigilator.**

Sample paper

SECTION A – ABSTRACT & SPATIAL THINKING

Q1. What is the decimal value of the binary number 101101?
A. 41
B. 43
C. 45
D. 47

Q2. A number N leaves remainder 4 when divided by 7. What remainder will 3N + 2 leave when divided by 7?
A. 0
B. 2
C. 4
D. 6

Q3. A 3 × 3 × 3 cube is painted on all outer faces and cut into unit cubes. How many unit cubes have exactly two painted faces?
A. 8
B. 12
C. 16
D. 20

Q4. The point (3, −2) is rotated 90° counter-clockwise about the origin. What are its new coordinates?
A. (−2, −3)
B. (2, 3)
C. (−2, 3)
D. (2, −3)

Q5. Reflecting a figure in a vertical line and then in a horizontal line through the same centre is equivalent to a rotation of:
A. 90°
B. 120°
C. 180°
D. 270°

Q6. A solid rectangular block is made of 3 × 2 × 2 unit cubes. How many unit-square faces are exposed on its surface?
A. 24
B. 28
C. 32
D. 36

Q7. A regular octagon is rotated about its centre. What is the smallest positive angle that leaves it unchanged?
A. 30°
B. 45°
C. 60°
D. 90°

Q8. A rectangular grid has 4 rows and 3 columns of small rectangles. How many grid intersection points are there?
A. 12
B. 16
C. 20
D. 24

Q9. Which of the following Capital letter has both horizontal vertical line of symmetry.
A. R
B. P
C. H
D. F

Q10. The point (5, −3) is reflected across the y-axis. What are its new coordinates?
A. (−5, −3)
B. (5, 3)
C. (−3, 5)
D. (3, −5)

SECTION B – PATTERN RECOGNITION

Q11. Find the next number: 2, 5, 12, 27, 58, ___.
A. 115
B. 119
C. 121
D. 123

Q12. Find the next number: 1, 2, 6, 15, 31, 56, ___.
A. 81
B. 86
C. 92
D. 96

Q13. These are binary numbers in order: 1, 10, 11, 100, 101, 110, 111, ___. What comes next?
A. 1000
B. 1001
C. 1010
D. 1111

Q14. Find the next number: 7, 14, 12, 24, 22, 44, 42, ___.
A. 62
B. 82
C. 84
D. 86

Q15. A rule maps n to $n^2 + n$. What is the output when n = 8?
A. 64
B. 70
C. 72
D. 80

Q16. The letters B, C, E, G, K, M, Q are at prime-numbered positions in the alphabet. Which letter comes next?
A. R
B. S
C. T
D. U

Q17. Find the next number: 128, 64, 32, 16, 8, ___.
A. 2
B. 4
C. 6
D. 12

Sample paper

Q18. Find the next number: 3, 7, 15, 31, ___.
A. 47
B. 55
C. 63
D. 71

Q19. Find the next number: 2, 6, 12, 20, 30, 42, ___.
A. 54
B. 56
C. 58
D. 60

Q20. Find the next pair: B2, E6, I12, N20, ___.
A. S28
B. T30
C. U30
D. U32

SECTION C – DECOMPOSITION, NUMBERS & DATA

Q21. What is the smallest number greater than 20 that leaves remainder 2 when divided by 5 and remainder 1 when divided by 3?
A. 21
B. 22
C. 27
D. 32

Q22. Five identical machines make 300 items in 4 hours. At the same rate, how many items will 3 machines make in 6 hours?
A. 240
B. 270
C. 300
D. 360

Q23. The mean of 12, 18, 20, 25 and x is 20. What is x?
A. 20
B. 23
C. 25
D. 27

Q24. Two boxes A and B contain balls in the ratio 5 : 3, with 64 balls altogether. If 8 balls are moved from A to B, what is the new ratio A : B?
A. 1 : 1
B. 3 : 2
C. 4 : 3
D. 5 : 4

Q25. An item marked ₹1600 gets a 15% discount. Then 5% tax is added to the discounted price. What is the final price?
A. ₹1360
B. ₹1400
C. ₹1428
D. ₹1440

Q26. A dataset of 1200 records is split 70% for training, 15% for validation and the rest for testing. How many records are used for testing?
A. 150
B. 180
C. 200
D. 240

Q27. In a class of 50 students, 28 like Coding, 24 like Robotics and 12 like both. How many like neither?
A. 8
B. 10
C. 12
D. 14

Q28. The numbers 4, 7, 9, 12, 15, 18, 21 have what median?
A. 9
B. 12
C. 15
D. 18

Q29. A dataset contains 800 rows. If 65% are training and 20% are validation, how many are test rows?
A. 100
B. 120
C. 140
D. 160

Q30. A 128 × 64 grayscale image uses 8 bits per pixel. Ignoring metadata, how many bytes are needed?
A. 4096
B. 8192
C. 16384
D. 65536

SECTION D – ALGORITHMIC THINKING & LOGICAL SEQUENCING

Q31. Apply this rule four times: if divisible by 3, divide by 3; otherwise if even, divide by 2; otherwise add 5. Start with 28. What is the result?
A. 2
B. 4
C. 6
D. 8

Q32. Start with x = 3. Repeat four times: x = 2x + 1. What is the final value of x?
A. 31
B. 47
C. 63
D. 65

Q33. Set S = 0. For n = 1 to 5: if n is even, add 2n to S; otherwise add n. What is the final value of S?
A. 18
B. 20
C. 21
D. 24

Q34. On a grid, a robot goes from (1,1) to (4,4) using only Right and Up moves. It may not pass through (2,2). How many shortest paths are possible?
A. 6
B. 8
C. 10
D. 12

B. 20

Sample paper

Q35. Four tasks obey: B is before A; A is before C; C is before D. Which task must be third?
A. A
B. B
C. C
D. D

Q36. To find the largest of a, b and c, an algorithm sets max=a, then uses 'if b < max, max=b'. What should replace '<' to fix it?

A. =
B. >
C. <=
D. !=

Q37. Set count = 0. For n from 1 to 10, increase count when n is divisible by 3. What is count at the end?

A. 2
B. 3
C. 4
D. 5

Q38. Start x = 20. Repeat: x = x/2 while x is even. What value stops the loop?
A. 2
B. 4
C. 5
D. 10

Q39. A shortest path needs exactly 4 Right moves and 2 Up moves. How many different shortest paths are possible?

A. 10
B. 12
C. 15
D. 18

Q40. An algorithm adds numbers 1 to 5 but accidentally resets sum=0 inside every loop iteration. What is the final displayed sum?

A. 0
B. 5
C. 10
D. 15

SECTION E – OLYMPIAD CHALLENGE

Q41. Which is the correct conceptual order of an AI project lifecycle?

A. Collect → Define → Reflect → Test
B. Define → Collect → Test → Reflect/Improve
C. Test → Collect → Define → Stop
D. Define → Test → Collect → Stop

Q42. An AI labels emails as 'spam' or 'not spam'. Which technique is this?

A. Classification
B. Regression
C. Clustering
D. Compression

Q43. An AI predicts the selling price of a house from its size, location and age. Which technique is most suitable?
A. Classification
B. Regression
C. Clustering
D. Translation

Q44. A system groups unlabeled news articles by similarity of topic. Which method best describes this?

A. Classification
B. Regression
C. Clustering
D. Encryption

Q45. A face-recognition dataset contains very few examples from one age group. What is the most important risk?

A. Larger files
B. Unfair or lower accuracy for that group
C. Slower internet
D. Inability to store numbers

Q46. What is $11101_2 + 1011_2$ in binary?

A. 100100_2
B. 101000_2
C. 101100_2
D. 110000_2

Q47. Which AI domain is mainly used to identify objects in photographs?
A. NLP
B. Computer Vision
C. Audio synthesis
D. Spreadsheet formulas

Q48. A game-playing agent improves by receiving rewards for good moves. This is:

A. Supervised learning
B. Reinforcement learning
C. Clustering
D. Compression

Q49. Why is a validation dataset useful?

A. To replace all training data
B. To tune/check the model before final testing
C. To hide labels permanently
D. To increase screen resolution

Q50. Which is the safest data practice?

A. Collect all possible personal data
B. Share raw student data publicly
C. Collect only necessary data with consent
D. Store passwords in plain text

Sample paper